Hospital Readmission Analysis

Reducing 30-day readmissions: patterns, risk factors and recommendations

Excel portfolio project (DACS08) | Prepared by Md Ziaul Haque

1. Executive summary

A hospital administration team saw readmissions spike, raising costs and raising questions about quality of care. This project analyses 48,911 unique patient encounters in Excel to find what is linked to readmission within 30 days and what the hospital can do about it.

Headline numbers

Measure	Result
Overall 30-day readmission rate	46.2% (22,609 of 48,911 encounters)
Average length of stay	4.4 days
Patients on diabetes medication	77.0% of encounters, and about 80% of all readmissions
Risk score range	26.2% readmitted at score 0 to 69.5% at score 6

Key findings

- **Prior emergency visits are the clearest warning sign.** Readmission climbs from 44.0% with no prior emergency visits to 87.1% with five or more.
- **Comorbidity matters.** Readmission rises from 22.6% (1 diagnosis) to 50.7% (9 diagnoses), and stay length more than doubles.
- **Diabetes drives the problem.** Diabetic patients are readmitted at 47.9% vs 40.5%, have the highest prior emergency and inpatient use, and account for roughly 80% of readmissions.
- **Medication changes are linked to higher risk.** Changed: 48.7% vs 44.1% with no change (statistically significant, $p < 0.001$).
- **Age 60 to 90 is the core population.** These ages hold 81.7% of encounters and the highest rates (46.7% to 48.2%).
- **Two medications look protective.** Among diabetic patients, X3 (-6.4 points) and X10 (-3.0 points) go with significantly lower readmission.
- **Race and gender show small gaps.** The two largest racial groups differ by under 1 point, and the gender gap is 1.3 points.

Top recommendations

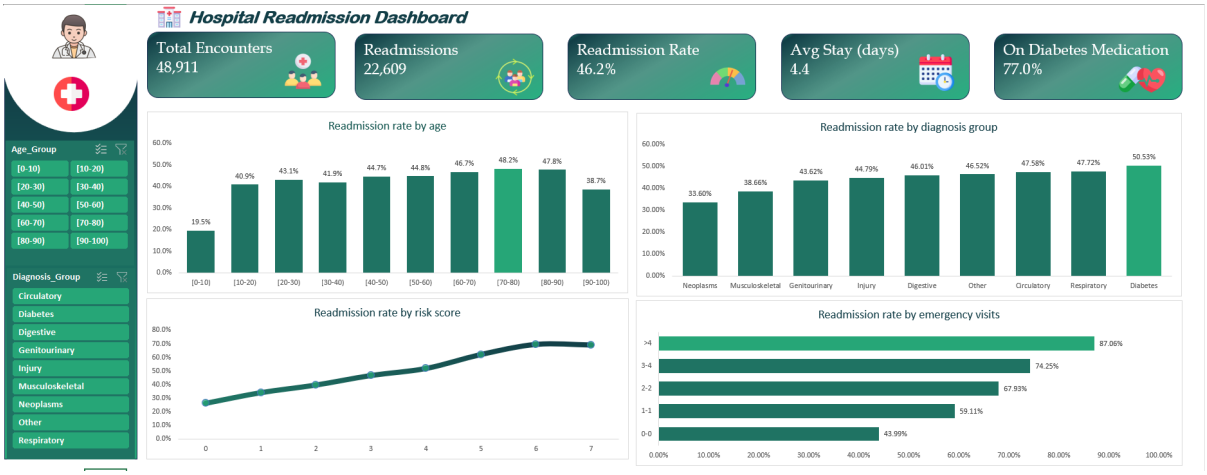
- Use a simple risk score at discharge: intensive follow-up for score 5 or more (about 9% of patients), an early follow-up call for score 3 to 4.
- Make diabetes the priority condition: medication review at discharge, education and an early outpatient visit.
- Add a pharmacist review for every patient whose medication was changed.

- Flag patients with any prior emergency visit, and treat three or more as high risk.
- Fix data gaps: weight (96.8% missing), specialty (49% unknown), and add dates, discharge disposition, cost and satisfaction fields.

The follow-up program in Advanced Q5 estimates that tiered follow-up could avoid roughly 564 to 1,233 readmissions. These figures are planning scenarios based on assumed reduction rates, not measured results.

2. Dashboard

An interactive Excel dashboard summarises the analysis. Five KPI cards (encounters, readmissions, readmission rate, average stay, share on diabetes medication) and four charts (age, diagnosis group, risk score, emergency visits) respond to slicers for age group and diagnosis group.



3. Business problem and objective

Item	Detail
Business problem	A spike in patient readmissions raises costs and may signal quality-of-care issues.
Role	Data analyst in the hospital administration department.
Objective	Find the patterns and factors linked to readmission within 30 days, and give actionable recommendations.
Target variable	readmitted (1 = readmitted within 30 days, 0 = otherwise).
Tool	Microsoft Excel: tables, formulas, pivot tables, charts, slicers.

4. Dataset and data dictionary

The raw file held 66,587 rows and 49 columns. After cleaning, the analysis dataset has 48,911 unique encounters. Columns by group:

Group	Columns	Notes
Identifiers	encounter_id, patient_id	Used only to check duplicates and join data.

Group	Columns	Notes
Demographics	race, gender, age	Age is stored in 10-year bands such as [60-70).
Stay and care	time_in_hospital, medical_specialty, num_lab_procedures, num_procedures, num_medications	Specialty is unknown for 49% of encounters.
Prior visits	number_outpatient, number_emergency, number_inpatient	Counts for the year before the stay.
Diagnosis	diag_1, diag_2, diag_3, number_diagnoses	diag_1 to diag_3 are ICD-9 style codes.
Medications and tests	X1 to X25	Anonymised. X1 and X2 hold test results (None, Norm, >200, >8, >7). X3 to X25 hold medication status (No, Steady, Up, Down).
Diabetes	change, diabetesMed	change was standardised to Yes/No.
Target	readmitted	1 = readmitted within 30 days.
Helper columns added	Readmit_Status, Total_Prior_Visits, Diagnosis_Group, Weight_Band, Risk_Score, Segment, Diabetes_Flag	Built during cleaning and analysis.

5. Approach and workbook structure

The work followed a fixed procedure, with every step documented before moving on:

- Setup: define the problem, keep the raw file untouched and work on a copy.
- Understand: check size, missing values, duplicates and data types.
- Clean: handle missing values, remove duplicates, standardise values, add helper columns.
- Analyse: one pivot table per question, each on its own sheet, with a log entry (method, result, insight, recommendation, limitation).
- Dashboard: charts and slicers on one sheet, with KPI cards linked to a pivot so they respond to slicers.
- Report: this document.

Workbook sheets: Raw file (kept separately), Clean_Data (table tbl_Hospital), Data_Quality, Project_Log, one sheet per question, KPI_Pivot, Dashboard_Calc and Dashboard.

6. Data understanding and cleaning

6.1 Data understanding

Check	Finding
Size at load	66,587 rows and 49 columns.

Check	Finding
Columns removed (49 to 46)	index (row counter), diag_4 (duplicate of diag_3), diag_5 (non-diagnostic decimal values).
Duplicates	17,676 exact duplicate rows (all columns identical), leaving 48,911 unique encounters.
Missing values	Question marks only: weight 96.8%, medical_specialty 49.0%, race 2.3%, diag_3 1.4%, diag_2 0.4%, diag_1 0.02%. No blank cells anywhere.
Data types	All numeric columns stored correctly. Categorical columns are text. diag_1 to diag_3 hold a mix of numeric and letter-coded values.

6.2 Cleaning log

Step	Action
Missing values	Replaced ? with Unknown in race (1,114), medical_specialty (23,983), diag_1 (11), diag_2 (172) and diag_3 (685).
Weight	Deleted the weight column from the working table (96.8% missing). Weight was later re-joined from the raw file for a small-sample analysis (Basic Q8, Medium Q8).
Standardising	change: Ch replaced with Yes to match the data dictionary (Yes/No).
Helper columns	Readmit_Status (label for the 1/0 target), Total_Prior_Visits (outpatient + emergency + inpatient), Diagnosis_Group (ICD-9 range grouping of diag_1, with Other at 18%).
Later additions	Weight_Band, Risk_Score, Segment and Diabetes_Flag were added for the weight analysis, risk score, segmentation and dashboard KPI cards.

7. Analysis and findings

Each question below follows the same format: question, method, result, insight, recommendation and limitation. All 25 questions from the brief are covered. Some can only be answered in part because the data lacks certain fields, and those are documented as limitations.

7.1 Basic questions (Q1 to Q10)

Basic Q1: Readmission by age group

Question: *What is the distribution of readmission rates across different age groups?*

Method	Pivot table (Q1_Age): age in rows, average of readmitted as the rate, plus encounter count.
Result	Overall rate 46.2%. Highest: 70-80 (48.2%), 80-90 (47.8%), 60-70 (46.7%). Lowest: 0-10 (19.5%, only 77 encounters, unreliable) and 90-100 (38.7%).
Insight	Readmission risk rises with age and peaks at 60-90, the groups that also hold 81.7% of all encounters.
Recommendation	Focus post-discharge follow-up (calls, early check-ups, medication review) on patients aged 60-90, where both risk and volume are highest.
Limitation	The data does not explain why the 90+ group is lower, so it is reported only.

Basic Q2: Length of stay by medical specialty

Question: *Which specialties have the longest and shortest average hospital stays?*

Method	Pivot table (Q2_Specialty): specialty in rows, average time_in_hospital and encounter count. Specialties with fewer than 100 encounters were filtered out as unreliable.
Result	Overall average 4.4 days. Longest: Physical Medicine and Rehabilitation (9.0), Psychiatry (6.7), Cardiovascular/Thoracic Surgery (6.0). Shortest: Obstetrics and Gynecology (3.1), Urology (3.2), Pediatrics and Radiologist (3.5).
Insight	Rehabilitation and psychiatry patients stay about twice as long as average, while procedure-based specialties stay shorter.
Recommendation	Plan beds, staffing and discharge planning around the long-stay specialties, and start discharge planning early for them.
Limitation	49% of encounters have an unknown specialty, so the picture covers only about half of the patients.

Basic Q3: Emergency visits and readmission

Question: *How does the number of emergency visits in the previous year correlate with readmission rates?*

Method	CORREL between number_emergency and readmitted, plus a pivot (Q3_Emergency) with visits grouped as 0, 1, 2, 3-4 and 5 or more.
Result	Correlation 0.109 (weak, positive). Readmission rises from 44.0% (0 visits) to 59.1% (1), 67.9% (2), 74.3% (3-4) and 87.1% (5 or more).

Insight	Each extra emergency visit raises readmission risk. The correlation looks weak only because 88.6% of patients have no emergency visits.
Recommendation	Flag every patient with one or more prior emergency visits for follow-up, and prioritise those with 3 or more (841 encounters, 74% or higher) for intensive care coordination.
Limitation	Association, not cause. Correlation captures only straight-line relationships, so it was confirmed with the grouped pivot.

Basic Q4: Diabetic patients and readmission

Question: *What proportion of patients with diabetes were readmitted within 30 days?*

Method	Pivot table (Q4_Diabetes): diabetesMed in rows, average of readmitted and encounter count.
Result	Diabetes medication: 47.94% readmitted (37,640 encounters). No diabetes medication: 40.50% (11,271 encounters). Overall: 46.22%.
Insight	Diabetic patients are readmitted more often, and they make up about 77% of encounters and roughly 80% of all readmissions.
Recommendation	Make diabetes patients the priority group for the readmission program: medication review at discharge, glucose-control education and an early follow-up visit.
Limitation	Diabetic is defined by diabetes medication (diabetesMed), not diagnosis codes, so patients with diabetes who take no medication fall into the No group.

Basic Q5: Medication change and readmission

Question: *Is there a significant difference in readmission rates between patients with changes in diabetic medication and those without?*

Method	Pivot table (Q5_MedChange) comparing the average of readmitted by the change column, followed by a two-proportion z-test.
Result	Changed: 48.69% (22,547 encounters). No change: 44.12% (26,364). Gap of 4.57 percentage points, z = 10.11, p = 4.97E-24.
Insight	Patients whose diabetic medication was changed are readmitted significantly more often, about 10% more than those with no change.
Recommendation	Review the medication-change policy. Add a pharmacist review at discharge and a follow-up call within a few days for every patient whose medication changed.
Limitation	Association, not cause. Patients with more complex conditions may be more likely to have changes, so the change itself may not drive readmission.

Basic Q6: Lab procedures and readmission

Question: *Investigate the relationship between the number of lab procedures during the stay and readmission rates.*

Method	CORREL between num_lab_procedures and readmitted, plus a pivot (Q6_LabProcedures) with lab procedures grouped in bands of 10.

Result	Correlation 0.037 (very weak, positive). Readmission rises from 42.0% (10-19 procedures) to 50.2% (70-79).
Insight	Patients who receive more lab procedures are readmitted somewhat more often, probably because they are sicker. The effect is gradual, not sharp.
Recommendation	Treat 60 or more lab procedures as a risk marker for follow-up. Review whether high lab volumes reflect needed or avoidable tests before changing any protocol.
Limitation	Association, not cause. Bands above 80 have too few encounters (812, 203 and 33) to read.

Basic Q7: Readmission by race and gender

Question: *How do readmission rates vary by race and gender?*

Method	Two pivot tables (Q7_RaceGender): average of readmitted by race (rows) and gender (columns), and a matching count of encounters.
Result	Race: Caucasian 47.0%, African American 46.1%, Hispanic 41.1%, Other 40.6%, Asian 37.0%, Unknown 31.5%. Gender: Female 46.8%, Male 45.5%.
Insight	The two largest racial groups (93.7% of encounters) have almost the same rate, and the gender gap is small (1.3 points). Lower rates for Asian, Hispanic and Other patients rest on much smaller groups.
Recommendation	The data does not support targeting a program by race or gender. Keep monitoring the Asian and Hispanic groups as volumes grow, and improve how race is recorded (2.3% is unknown).
Limitation	The gender gap was not tested for significance, and cells under about 500 encounters are unreliable. Lower rates in small groups could reflect care received elsewhere.

Basic Q8: Weight categories

Question: *What is the distribution of patients across weight categories, and how does this relate to readmission rates?*

Method	Missing-value check on weight, then weight and encounter_id were taken from the raw file, de-duplicated and joined back to the table. Pivot (Weight_Pivot) of readmission rate and count by weight band for patients with a recorded weight.
Result	96.8% of weights are missing (47,354 of 48,911), leaving 1,557 usable records. Readmission: 53.6% (50-75), 60.2% (75-100), 57.7% (100-125), with no clear trend.
Insight	Weight data is too incomplete to support conclusions about all patients. The small weighed group has a much higher readmission rate (57.6% vs 46.2%), so it is not representative.
Recommendation	Make weight a required field at admission so a proper analysis becomes possible.
Limitation	Indicative only: a 3.2% sample from a different kind of patient. Bands with fewer than 100 patients were not interpreted. Weight units are not stated in the data dictionary.

Basic Q9: Medications and length of stay

Question: *Evaluate the impact of the number of medications prescribed on the length of hospital stay.*

Method	CORREL and a simple linear regression (SLOPE, INTERCEPT, RSQ) of time_in_hospital on num_medications.
Result	Correlation 0.465, slope 0.171 days per medication, intercept 1.67, R-squared 0.216. Predicted stay: about 3.4 days for 10 medications and 6.8 days for 30.
Insight	Patients on more medications stay longer, by about 0.17 days for each extra medication. Medication count explains 21.6% of the variation in stay length.
Recommendation	Use medication count at admission as an early flag for likely long stays in bed and staffing planning. Add a pharmacist review for patients on high medication counts.
Limitation	Association, not cause: sicker patients receive more medications and also stay longer. A straight line explains only 21.6% of the variation.

Basic Q10: Outpatient visits and readmission

Question: *Assess the frequency and impact of outpatient visits in the year prior to the current stay on readmission rates.*

Method	CORREL between number_outpatient and readmitted, plus a pivot (Q10_Outpatient) with visits grouped as 0, 1, 2, 3-4 and 5 or more.
Result	Correlation 0.083 (very weak, positive). 83.4% of encounters had no prior outpatient visit (43.8% readmitted). With any outpatient visit, readmission is 57.6% to 61.4%.
Insight	Prior outpatient contact marks higher-risk patients. The risk jumps at the first visit and then levels off, and more visits did not go with fewer readmissions.
Recommendation	Treat any prior outpatient visit as a risk flag at discharge. Review whether outpatient follow-up for these patients reaches the right issues.
Limitation	Association, not cause. Patients who visit outpatient clinics are likely already sicker, and visit quality is not recorded.

7.2 Medium questions (Q1 to Q10)

Medium Q1, Q3, Q6 and Q10: not answerable with this data

Question	Topic	Why it cannot be answered	Data that would fix it
Q1	Readmission trend over time	No date, month, or admission and discharge date column.	Admission and discharge dates.
Q3	Discharge disposition and readmission	No discharge disposition column exists.	Discharge disposition codes (home, transfer and so on).
Q6	Outcomes beyond readmission (mortality)	Only the 30-day readmission flag exists, with no mortality or follow-up outcome.	Mortality and post-discharge outcome records.

Question	Topic	Why it cannot be answered	Data that would fix it
Q10	Patient satisfaction and readmission	No satisfaction scores exist.	Survey data linked to encounters.

Recommendation for all four: add these fields to the hospital data collection so the analyses become possible.

Medium Q2: Comorbidity, stay and readmission

Question: *Analyze the effect of comorbid conditions (multiple diagnoses) on length of stay and readmission rates.*

Method	Pivot table (MQ2_Comorbidity) using number_diagnoses as the comorbidity score, with average stay, readmission rate and encounter count.
Result	From 1 to 9 diagnoses, average stay rises from 2.3 to 5.0 days and readmission from 22.6% to 50.7%. Patients with 6 to 9 diagnoses are readmitted at 49.0% against 35.7% for 1 to 5.
Insight	Each additional diagnosis goes with longer stays and higher readmission risk. Comorbidity is one of the strongest risk signals found.
Recommendation	Flag patients with 6 or more diagnoses at admission for comorbidity-focused discharge planning, and plan beds and staffing around their longer stays.
Limitation	Number of diagnoses counts coded conditions, not severity. Nearly half of encounters sit at exactly 9, which suggests a recording cap, and the 10 to 16 groups (52 encounters) are too small to read.

Medium Q4: Demographics and interventions

Question: *Explore the relationship between demographics (race, gender, age) and the interventions received (lab procedures, medications).*

Method	Three pivots (MQ4_Demographics) of average lab procedures, average medications and count by gender, race and age, plus a two-sample t-test between women and men.
Result	Gender: no real difference in labs (43.2 vs 43.0, $p = 0.277$) and a 0.2 medication gap (16.1 vs 15.9, $p = 0.0016$). Race: medications range from 13.1 (Asian) to 16.3 (Caucasian). Age: medications peak at 17.2 for ages 60-70 while labs stay flat.
Insight	Treatment intensity follows age, not gender. Race differences exist in medication counts but may reflect age and health mix, not unequal care.
Recommendation	No gender-based intervention is needed. Review prescribing for Asian and Hispanic patients to check whether lower counts reflect age or health mix or a gap in care. Run polypharmacy reviews for patients aged 60 and over.
Limitation	Only gender was tested for significance. Race and age gaps are descriptive and not adjusted for each other or for diagnosis. Asian (300) and 0-10 (77) groups are small.

Medium Q5: Prior visits and primary diagnosis

Question: *Examine the association between prior inpatient, outpatient and emergency visits and the type of primary diagnosis.*

Method	Pivot table (MQ5_Diagnosis) using the Diagnosis_Group helper column (ICD-9 ranges of diag_1), with the average of each visit type and encounter count.
Result	Diabetes has the highest prior use (0.90 inpatient, 0.39 emergency, 1.72 total visits against 1.22 overall). Neoplasms and musculoskeletal diagnoses are lowest (0.81 and 0.86 total). Outpatient use is flat across groups.
Insight	Diabetes patients return to hospital and emergency care far more often than other groups, which supports treating diabetes as the priority condition for prevention.
Recommendation	Build a preventive diabetes care pathway (outpatient management, education, early follow-up), and plan emergency and inpatient capacity with diabetes as a high-use group.
Limitation	Diagnosis groups use broad ICD-9 ranges, and Other (18%) mixes letter-coded and unknown diagnoses. Averages show association, not cause, and do not measure severity.

Medium Q7: Prescribing patterns by specialty

Question: *Assess the variation in prescribing patterns (X1 to X25) across medical specialties and their impact on readmission rates.*

Method	Prescribing frequency table for X3 to X25 (MQ7_Prescribing), readmission with and without each medication, and a pivot (MQ7_Specialty) of X20 status by the five largest specialties as a share of each row. X1 and X2 are excluded because they hold test results.
Result	Only 7 of 23 medications are used by 5% or more of encounters. X20 goes with +4.4 points higher readmission and X3 with -3.8. X20 prescribing runs from 64.0% (Emergency/Trauma) to 44.8% (Cardiology), with dose changes in 41.4% vs 16.5% of encounters.
Insight	Prescribing differs a lot between specialties, and two medications (X20, X3) show readmission gaps worth following up. Emergency/Trauma adjusts doses far more often than other specialties.
Recommendation	Review X20 prescribing and dose-change practice in Emergency/Trauma, study why X3 patients have lower readmission, and standardise medication-change protocols across specialties.
Limitation	Medication names are anonymised, so findings cannot be tied to specific drugs. The all-patient gaps are not adjusted for patient mix. Only the five largest specialties are compared, and cluster analysis was not used.

Medium Q8: Weight and diagnosis together

Question: *Analyze the relationship between weight categories and specific diagnosis on readmission rates.*

Method	Two pivots (Weight_Pivot): readmission rate and encounter count by Diagnosis_Group and weight band, for the three bands with 100 or more patients (50-75, 75-100, 100-125), covering 1,393 weighed patients.
Result	Diabetes has the highest readmission (74.0%, and 82.9% at 75-100) and Neoplasms the lowest (39.1%). Weight bands differ by only about 7 points overall, with no consistent pattern across diagnoses.
Insight	Diagnosis, especially diabetes, predicts readmission far better than weight. Weight adds a small, inconsistent effect.
Recommendation	Do not build weight-based programs on this evidence. Prioritise diabetes patients, and record weight for every patient so this can be re-tested properly.
Limitation	Only 3.2% of patients have a weight, and they differ from the rest (57.6% vs 46.2% readmitted). 8 of 27 cells have fewer than 30 patients. No significance tests were run.

Medium Q9: Diabetic medication effectiveness

Question: *Explore the effectiveness of different diabetic medications (X variables) on preventing readmissions among diabetic patients.*

Method	For the 7 usable medications, a comparison table (MQ9_DiabeticMeds) of readmission with and without each medication among diabetic patients only (37,640 encounters), with a two-proportion z-test and p-value for each.
Result	X3 (-6.4 points, p = 2.1E-27) and X10 (-3.0 points, p = 6.9E-05) go with significantly lower readmission. X20 shows +1.2 points (borderline, p = 0.038). X7, X9, X12 and X13 show no significant difference.
Insight	Two medications (X3 and X10) are associated with fewer readmissions among diabetic patients. The apparent X20 effect in Medium Q7 was mostly a diabetic vs non-diabetic difference (+4.4 shrinks to +1.2).
Recommendation	Investigate prescribing protocols and patient mix for X3 and X10, and consider promoting their use where clinically suitable. Review X20 practice, but do not treat it as harmful on this evidence.
Limitation	Medication names are anonymised. These are associations, not proof of effectiveness: patients given X3 or X10 may be healthier or younger. Not adjusted for age, diagnosis or severity, and 7 tests raise the chance of a false positive.

7.3 Advanced questions (Q1 to Q5)

Advanced Q1: Risk score for high-risk patients

Question: *Develop a predictive model to identify patients at high risk of readmission.*

Method	A points-based risk score (Risk_Score, 0 to 7) built from factors found earlier: emergency visits (0 to 2 points), outpatient visits, 6 or more diagnoses, diabetes medication, medication change and age 60 to 89. Checked with a pivot (Q_Advanced1_Risk).
Result	Readmission rises from 26.2% (score 0) to 69.5% (score 6). Bands: low (0-2) 37.2% across 17,199 patients, medium (3-4) 49.1% across 27,257, high (5-7) 63.3% across 4,455.
Insight	Simple, known risk factors rank patients well: the highest band is readmitted about 1.7 times as often as the lowest. The score is useful for triage but separates patients only moderately.
Recommendation	Use the score at discharge: score 5 or more gets intensive follow-up (about 9% of patients), and score 3 to 4 gets an early follow-up call. Review the weights with clinicians.
Limitation	An Excel rule-based score, not a fitted model. It was checked on the same data used to build it, with no test set or accuracy measure such as AUC. Logistic regression or decision trees in Python are the natural next step.

Advanced Q2: Patient segmentation

Question: *Create a patient segmentation based on utilisation patterns and outcomes.*

Method	Rule-based segments from Total_Prior_Visits and number_emergency, then a pivot (Q_Advanced2_Segments) of readmission, encounters, average stay and average medications.
Result	Readmission: no prior visits 36.2% (26,716), other prior visits 55.3% (15,161), emergency users 57.6% (3,894), high utilisers (5 or more visits) 73.9% (3,140). Stay and medication use differ little.
Insight	Prior hospital use separates patients clearly by risk. High utilisers are a small group with by far the highest rate, while most readmissions in absolute numbers come from the large lower-risk groups.
Recommendation	Tier the follow-up: intensive support for high utilisers (case manager and a call within 48 hours), an early visit for emergency users, and standard follow-up with reminders for the rest.
Limitation	Rule-based segments with thresholds chosen by us, not statistical clustering such as K-means. Segments show association, not cause.

Advanced Q3 and Q4: partly covered

Question	What the data can tell us	What is missing	Recommendation
Q3: Financial impact, avoidable vs unavoidable readmissions	22,609 of 48,911 encounters (46.2%) were readmitted. Each 1-point drop in the rate avoids about 489 readmissions.	No cost data, and no clinical criteria or reason codes to separate avoidable from unavoidable readmissions.	Multiply 489 by the hospital average cost per readmission to estimate savings per point. Add cost and reason fields to the data.
Q4: Social determinants of health	Race analysed in Basic Q7: Caucasian 47.0%, African American 46.1%, Hispanic 41.1%, Other 40.6%, Asian 37.0%. The two largest groups are nearly identical.	No socioeconomic data (income, insurance, education, housing) and no external dataset.	Collect insurance type, area income and housing status at admission, then repeat. Do not target interventions by race on this evidence.

Advanced Q5: Follow-up program and estimated impact

Question: *Design a post-discharge follow-up program based on the analysis and assess its potential impact.*

Tier	Who	Actions	Evidence
1: Intensive	Risk score 5 to 7 (4,455 patients, about 9%)	Case manager. Call within 48 hours. Visit within 7 days. Pharmacist medication review. Diabetes education if on diabetes medication.	63.3% readmitted (Advanced Q1). 5 or more emergency visits: 87.1% (Basic Q3). High utilisers: 73.9% (Advanced Q2).
2: Early contact	Risk score 3 to 4 (27,257 patients, about 56%)	Automated call or text within 3 to 5 days. Check instructions are understood and prescriptions filled.	49.1% readmitted, and the band holds the most readmissions (about 13,400).
3: Standard	Risk score 0 to 2 (17,199 patients)	Normal discharge process.	37.2% readmitted, the lowest band.
All tiers	Medication changed	Pharmacist review before discharge.	48.7% vs 44.1% (Basic Q5).
All tiers	Diabetes patients	Diabetes management pathway: education and early outpatient visit.	47.9% readmitted, about 80% of all readmissions (Basic Q4, Medium Q5).
System	Hospital management	Review prescribing of X3 and X10. Fix data gaps (weight, specialty, cost, discharge).	Medium Q9, Basic Q8, data quality checks.

Estimated impact (planning scenarios, not results)

The reduction percentages are assumptions. This dataset can show how many readmissions each assumption would avoid, but not how well the program works.

Scenario	Readmissions avoided	New overall rate
Tier 1 cuts readmissions by 10%	about 282	45.6% (from 46.2%)
Tier 1 cuts readmissions by 20%	about 564	45.1%
Tier 1 cuts readmissions by 30%	about 846	44.5%
Tier 1 at 20% and Tier 2 at 5%	about 1,233	43.7%

Recommendation: pilot the program on the highest-risk band and the medication-change and diabetes groups for a few months, measure the actual reduction, and expand if results hold. Limitation: the reduction rates are assumed, and the data has no cost, discharge or follow-up information.

8. Consolidated recommendations

#	Area	Recommendation
1	Risk triage	Score every patient at discharge and tier follow-up by score.
2	Diabetes	Create a diabetes readmission pathway: medication review, education, early outpatient visit.
3	Medication safety	Pharmacist review for medication changes and for patients on high medication counts.
4	Prior use	Flag any prior emergency visit, and escalate three or more.
5	Capacity planning	Use age, comorbidity and specialty patterns to plan beds and staffing, especially rehabilitation and psychiatry.
6	Prescribing	Review X20 practice in Emergency/Trauma and study X3 and X10 for lower readmission.
7	Data quality	Make weight mandatory, reduce unknown specialty and race, and add dates, discharge disposition, cost, satisfaction and socioeconomic fields.

9. Limitations

- All findings are associations, not proof of cause. Sicker patients often receive more tests, medications and visits, and also return more often.
- Medication names are anonymised (X1 to X25), so drug-level conclusions cannot be tied to specific drugs.
- No date, discharge, mortality, cost, satisfaction or socioeconomic fields exist, so six brief questions could not be answered and two only in part.
- Weight is 96.8% missing, and the weighed group is not representative. Specialty is unknown for 49% of encounters.
- The risk score and segments are rule-based Excel tools, checked on the same data used to build them, with no held-out test or accuracy measure. Logistic regression or decision trees in Python would be the natural next step.
- Several comparisons rest on small groups, which are flagged in each answer and not interpreted.